PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T31 Classified

Cross-session • chapter-organised candidate

10 classified items • 43 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T31 • 10 items • 43 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Coordinate Geometry

Unit 1 • 43 marks • 10 item(s)

June 2023 | 4MA1-2H Q24 | 6 marks
Kite or symmetric vertex coordinates

June 2022 | 4MA1-2H Q22 | 7 marks
Kite or symmetric vertex coordinates

January 2020 | 4MA1-1H Q1 | 2 marks
Midpoint from two endpoints

January 2020 | 4MA1-1H Q22 | 6 marks
Coordinates from isosceles and gradient constraints

January 2021 | 4MA1-2H Q18 | 4 marks
Rectangle or parallelogram missing vertex

May-Nov 2020 | 1H Q22 | 4 marks
Read a plotted coordinate pair

January 2023 | 1H Q23 | 6 marks
Read a plotted coordinate pair

January 2023 | 2H Q6 | 3 marks
Rectangle or parallelogram missing vertex

November 2025 | 1H Q20 | 3 marks
Read a plotted coordinate pair

November 2025 | 2H Q1 | 2 marks
Rectangle or parallelogram missing vertex

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 28

4MA1-2H Q24

ORIGINAL QUESTION
6 marks

24 *ABCD* is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.......................... , ...)

Worked Solution 28

Coordinates of the fourth vertex of a kite

MODULAR UNIT 1
6 marks

Find AC

$$m_{AC} = \frac{-5-10}{4-(-2)} = -\frac{5}{2}, \quad y = 5 - \frac{5}{2}x.$$

Use the perpendicular diagonal

$$m_{BD} = \frac{2}{5}, \quad y - 4 = \frac{2}{5}\left(x + \frac{27}{5}\right).$$

Find the diagonal intersection

$$AC \cap BD = \left(-\frac{2}{5}, 6\right).$$

Reflect B in AC

$$D = 2\left(-\frac{2}{5}, 6\right) - \left(-\frac{27}{5}, 4\right) = \left(\frac{23}{5}, 8\right).$$

Final answer

$$D = (4.6, 8)$$

Original question 28

4MA1-2H Q22 Q22

ORIGINAL QUESTION
7 marks

22 *ABCD* is a kite, with diagonals *AC* and *BD*, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates (−3, 4)

The diagonals of the kite intersect at the point *M* with coordinates (0, 2)

Given that $AB = AD = 6.5$ cm and the x coordinate of *B* is positive,

find the coordinates of the points *B* and *D*.

(.......................... , ...)

(.......................... , ...)

Worked solution 28

Chapter: Coordinate Geometry

MODULAR UNIT 1
7 marks

Q22 – Chapter: Coordinate Geometry

Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordinates

Method
The gradient of AM is (2 - 4)/(0 - (-3)) = -2/3.
Why: Use the two given coordinates.

Method
The kite diagonals are perpendicular, so BD has gradient 3/2 and equation y = (3/2)x + 2.
Why: Use the negative reciprocal and point M.

Method
For B or D, (x + 3)^2 + (y - 4)^2 = 6.5^2.
Why: Each point is 6.5 units from A.

Method
Substitute y = (3/2)x + 2: (x + 3)^2 + ((3/2)x - 2)^2 = 42.25.
Why: Combine the line and distance conditions.

Method
Expanding gives 3.25x^2 + 13 = 42.25.
Why: Expand and collect; the linear x terms cancel.

Method
Thus x^2 = 9, so x = 3 or x = -3; the corresponding y-values are 6.5 and -2.5.
Why: Solve the quadratic and substitute into the line.

Method
Since B has positive x-coordinate, B = (3, 6.5) and D = (-3, -2.5).
Why: Use the final labelling condition.

Final answer
B: B = (3, 6.5)

Final answer
D: D = (-3, -2.5)

1 The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB.

(.............................. , ...)
(2)

Worked solution

January 2020 | Question 1 | 2 marks

Family: Use the midpoint relationship • Variant: Midpoint from two endpoints

Method / working

1. Midpoint $x=(5+13)/2=9$.
2. Midpoint $y=(-4+1)/2=-1.5$.

Final answer: $(9, -1.5)$

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$
J is the point with coordinates $(j, 15)$ where $j < 0$
K is the point with coordinates $(6, k)$

M is the midpoint of JK.
The gradient of HM is 2

Find the value of j and the value of k.

$j =$ ....

$k =$ ....

Worked solution

January 2020 | Question 22 | 6 marks

Family: Determine coordinates from geometric information • Variant: Coordinates from isosceles and gradient constraints

Method / working

1. $M=((j+6)/2,(15+k)/2)$.
2. Gradient HM=2 gives $(k+13)/(j+14)=2$, so $k=2j+15$.
3. $JK^2=80$ gives $(6-j)^2+(k-15)^2=80$.
4. Substitute k: $5j^2-12j-44=0$.
5. The condition $j<0$ selects $j=-2$.
6. Then $k=2(-2)+15=11$.

Final answer: $j=-2$, $k=11$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates (–4, –2)
B has coordinates (1, 10)
C has coordinates (19, a)
D has coordinates (b, c)

(a) Work out the value of a, the value of b and the value of c.

a = ...

b = ...

c = ...

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

.... cm

(3)

Worked solution

January 2021 | Question 18 | 4 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Method / working

1. AB=(12,5), so b=14 from the rectangle translation.
2. Gradient AB=12/5, hence gradient BC=-5/12.
3. Use C=(19,a) to obtain a=2.5 and then D=(14,-9.5).
4. State all three coordinates values.

Final answer: a=2.5, b=14, c=-9.5

22 *ABCD* is a rhombus.

The diagonals, *AC* and *BD*, intersect at the point *M*.
The coordinates of *M* are $(6, -11)$

The points *A* and *C* both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through *B* and *D* intersects the *y*-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

Answer reference | May-Nov 2020 | Question 22

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

22	$y=-\frac{7}{2}x(+10)$ **or** (gradient =) $-\frac{7}{2}$		4	B1	for correct gradient which may be seen in an equation. Condone $-\frac{7}{2}x$
	$'-\frac{7}{2}'m=-1$ **or** $(m=)'\frac{2}{7}'$			M1	ft their gradient for use of $m_1 \times m_2 = -1$
	$-11='\frac{2}{7}'\times 6+c$ **or** $y--11='\frac{2}{7}'(x-6)$ oe			M1	ft dep on M1
		$\left(0, -\frac{89}{7}\right)$		A1	accept $\left(0, -12\frac{5}{7}\right)$ must be exact values
					Total 4 marks

Worked solution

May-Nov 2020 | Question 22 | 4 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Method
In a rhombus, the diagonals are perpendicular and bisect each other.
The diagonal AC lies on
2y+7x=20
2y=-7x+20
y=-frac72x+10
So the gradient of AC is
-frac72
Therefore the gradient of BD is
frac27
The line BD passes through
M=(6,-11)
So
y+11=frac27(x-6)
To find where this line intersects the y-axis, put x=0:
y+11=frac27(0-6)
y+11=-(12)/(7)
y=-11-(12)/(7)
y=-(89)/(7)
Answer: displaystyle (0,-(89)/(7))

23 $ABCD$ is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D

The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

January 2023 | 1H | Question 23 | 6 marks

Chapter OL-T31: Coordinate Geometry

....

(Total for Question 23 is 6 marks)

Answer reference | January 2023 | Question 23

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

23	$y=\frac{3}{5}x\left(+\frac{6}{5}\right)$ or $y=0.6x(+1.2)$ **or** (gradient =) $\frac{3}{5}$ or 0.6 oe		6	M1 for correct gradient which may be seen in an equation. Condone $\frac{3}{5}x$ or $0.6x$
	$k=-2$			B1 for $k=-2$
	$\left(\frac{"-2"+8}{2},\frac{1+7}{2}\right)$ oe or (3, 4)			M1 for finding the midpoint (use of their k where $k<0$)
	$"\frac{3}{5}"m=-1$ **or** $(m=)"-\frac{5}{3}"$			M1ft their gradient for use of $m_1\times m_2=-1$ Allow $-\frac{5}{3}=-1.67$ or better
	$"4"="-\frac{5}{3}"\times"3"+c$ or $c=9$ or $y-"4"="-\frac{5}{3}"(x-"3")$			M1 dep on M3
	Working required	$5x+3y=27$		A1 allow equation in any form where p, q and r are integers
				Total 6 marks

Worked solution

January 2023 | Question 23 | 6 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Official final mark-scheme pages are embedded immediately after this question.

6 The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$
The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j =$ ....

$k =$ ....

(Total for Question 6 is 3 marks)

Answer reference | January 2023 | Question 6

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Question	Working	Answer	Mark	Notes
6	$[k=]\ \frac{6+17}{2}$ **or** $[k=]\,6+\frac{17-6}{2}$ oe **or** $[j=]\ 4+2(15-4)$ **or** $[j=]\ 15+(15-4)$ **or** $\frac{4+j}{2}=15$ oe		3	M1
	Correct answers score full marks (unless from obvious incorrect working) *1 correct answer will score M1A1 and both will score M1A1A1*	26		A1
		11.5		A1 oe eg $\frac{23}{2}$ both answers the wrong way round scores M1A1 unless the correct answers are clearly labelled in working space
				Total 3 marks

Worked solution

January 2023 | Question 6 | 3 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Official final mark-scheme pages are embedded immediately after this question.

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

....

(Total for Question 20 is 3 marks)

Worked solution

November 2025 | Question 20 | 3 marks

Family: Read and plot Cartesian coordinates • Variant: Read a plotted coordinate pair

Method
Line L has equation
y=4x+7
So its gradient is
4
Line M is perpendicular to L, so its gradient is
-frac14
Line M passes through (8,1).
y-1=-frac14(x-8)
y-1=-frac14x+2
y=-frac14x+3
Answer: displaystyle y=-frac14x+3

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

1 The point P has coordinates (3, 4)
The point Q has coordinates (9, 16)

M is the midpoint of the line PQ

Find the coordinates of M

(........................ , ...)

(Total for Question 1 is 2 marks)

Worked solution

November 2025 | Question 1 | 2 marks

Family: Determine coordinates from geometric information • Variant: Rectangle or parallelogram missing vertex

Solution
P=(-3,0),qquad Q=(9,16)
The midpoint is
((-3+9)/(2),(0+16)/(2))
=(3,8)
Answer: (3,8).

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION